Huawei Access Node Optical Transport Failure ALARM_7030 SOP

Node: YEL-ACC-52 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_7030

Description

The failure domain of Optical Transport Failure on Huawei Access Nodes (Role: Access) occurs due to the degradation of SFP/XFP modules, fiber cuts, or Rx/Tx power loss. This failure is a result of the physical degradation of the fiber optic link or the optical transceiver module. The Huawei Access Node uses the Optical Transport Network (OTN) protocol to forward data between nodes. OTN is a high-speed data transmission system that uses optical fibers to transmit data over long distances. When a SFP/XFP module fails or a fiber cut occurs, the OTN protocol detects the failure and triggers an alarm, resulting in the ALARM_7030 critical alarm. The failure of an optical transceiver module or a fiber cut can be caused by various factors such as high temperature, high humidity, or physical damage to the fiber optic cable. In order to troubleshoot and resolve this issue, it is essential to investigate the physical layer of the network and the optical transceiver modules.

Pre-requisite Checks

- Check: Verify that the Yelahanka Sub-station is accessible and all necessary equipment is available., Status: Completed

- Check: Verify that the redundant link is operational., Status: Completed

- Check: Verify that the power supply to the access node is stable., Status: Completed

Potential Root Causes

Cause	Probability
SFP/XFP module degradation	High
Fiber cut or damage	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display device transceiver, Expected Output: Transceiver Information, Description: This command displays the transceiver information, including the type, model, and status of the transceiver module. It helps to identify if the transceiver module is degraded or faulty.

Command: display fiber info, Expected Output: Fiber Information, Description: This command displays the fiber information, including the type, length, and status of the fiber optic cable. It helps to identify if the fiber is cut or damaged.
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```
Command: display otu info, Expected Output: OTU Information, Description: This command displays the OTU information, including the type, rate, and status of the OTU. It helps to identify if the OTU is functioning correctly.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace transceiver slot with , Description: This step involves replacing the degraded SFP/XFP module with a new one. The new transceiver module should be inserted into the same slot where the old one was removed.

Step 2: Step: Repair or replace the fiber optic cable, Command: repair fiber , Description: This step involves repairing or replacing the damaged or cut fiber optic cable. The new fiber cable should be installed in the same location as the old one.

Step 3: Step: Adjust the Rx/Tx power settings, Command: adjust tx power , Description: This step involves adjusting the Rx/Tx power settings to ensure that the signal strength is optimal. The new power setting should be set to a value that is within the recommended range for the transceiver module.

Post-Check Verification

```
Command: display device transceiver, Expected Output: Transceiver Information, Description: This command is used to verify that the transceiver module is functioning correctly after the resolution step.

Command: display fiber info, Expected Output: Fiber Information, Description: This command is used to verify that the fiber optic cable is functioning correctly after the resolution step.

Command: display otu info, Expected Output: OTU Information, Description: This command is used to verify that the OTU is functioning correctly after the resolution step.
```

Roll-back Procedures

- Step: Restore the original transceiver module, Command: replace transceiver slot with , Description: This step involves restoring the original transceiver module that was replaced during the resolution step.

- Step: Restore the original fiber optic cable, Command: restore fiber , Description: This step involves restoring the original fiber optic cable that was replaced during the resolution step.

- Step: Restore the original Rx/Tx power settings, Command: adjust tx power , Description: This step involves restoring the original Rx/Tx power settings that were adjusted during the resolution step.